WORKED SOLUTION

STEP II 1988, Question 1

Pure • Partial Fractions and Series

STEP II 1988, Question 1

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/88-s2-q1

The key idea

Split f into partial fractions, so that everything comes down to expanding two simple fractions, $\frac{1}{x-1}$ and $\frac{1}{x-2}$, as geometric series. The only real decision is *which way round* to expand each one. A geometric series $\sum r^n$ converges only when $|r| < 1$, so for each fraction we pull out whichever of the two terms in the denominator is larger in size. That choice changes as $|x|$ passes 1 and then 2, and this is why there are three different expansions.

Solution

Partial fractions

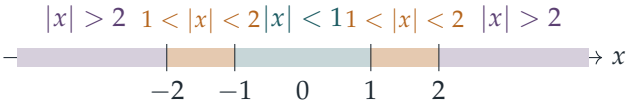
Write $\dfrac{1}{(x-1)(x-2)} = \dfrac{A}{x-1} + \dfrac{B}{x-2}$. The cover-up rule gives $A = \frac{1}{1-2} = -1$ and $B = \frac{1}{2-1} = 1$, so

$$\mathrm{f}(x) = \frac{1}{x-2} - \frac{1}{x-1}.$$

The two building blocks we need are the geometric series

$$\frac{1}{1-r} = \sum_{n=0}^{\infty} r^n \qquad (|r| < 1),$$

used with r equal to one of x, $\frac{x}{2}$, $\frac{1}{x}$ or $\frac{2}{x}$, whichever has modulus less than 1 in the range we are working in.



The range $|x| < 1$

Here $|x| < 1$ and also $|\frac{x}{2}| < \frac{1}{2} < 1$, so both fractions can be expanded in powers of x:

$$-\frac{1}{x-1} = \frac{1}{1-x} = \sum_{n=0}^{\infty} x^n,$$
$$\frac{1}{x-2} = -\frac{1}{2}\cdot\frac{1}{1-\frac{x}{2}} = -\frac{1}{2}\sum_{n=0}^{\infty}\left(\frac{x}{2}\right)^n.$$

Adding,

$$f(x) = \sum_{n=0}^{\infty} x^n - \frac{1}{2}\sum_{n=0}^{\infty}\left(\frac{x}{2}\right)^n \qquad (|x| < 1),$$

as required.

The range $1 < |x| < 2$

The second fraction is unchanged: we still have $|\frac{x}{2}| < 1$, so $\frac{1}{x-2} = -\frac{1}{2}\sum_{n\geq 0}(\frac{x}{2})^n$ exactly as before.

Ask yourself Why can't we keep using $\sum x^n$ for the first fraction?

Because $|x| > 1$ now, so the terms x^n do not tend to zero and $\sum x^n$ diverges. Instead we take out the larger term, x, from the denominator, which leaves a ratio $\frac{1}{x}$ with $|\frac{1}{x}| < 1$:

$$-\frac{1}{x-1} = -\frac{1}{x}\cdot\frac{1}{1-\frac{1}{x}} = -\frac{1}{x}\sum_{n=0}^{\infty} x^{-n} = -\sum_{n=0}^{\infty} x^{-(n+1)} = -\sum_{n=1}^{\infty} x^{-n},$$

where the last step just relabels $n+1$ as n. Adding the two pieces,

$$f(x) = -\sum_{n=1}^{\infty} x^{-n} - \frac{1}{2}\sum_{n=0}^{\infty}\left(\frac{x}{2}\right)^n \qquad (1 < |x| < 2),$$

as required.

The range $|x| > 2$

The first fraction keeps its expansion from the previous range, since $|\frac{1}{x}| < \frac{1}{2} < 1$. But now $|\frac{x}{2}| > 1$, so the second fraction must be turned round in the same way, taking out x and leaving the ratio $\frac{2}{x}$, which has $|\frac{2}{x}| < 1$:

$$\frac{1}{x-2} = \frac{1}{x}\cdot\frac{1}{1-\frac{2}{x}} = \frac{1}{x}\sum_{n=0}^{\infty}\left(\frac{2}{x}\right)^n = \sum_{n=0}^{\infty} 2^n x^{-(n+1)} = \sum_{n=1}^{\infty} 2^{n-1}x^{-n}.$$

Hence, for $|x| > 2$,

$$f(x) = \sum_{n=1}^{\infty} 2^{n-1}x^{-n} - \sum_{n=1}^{\infty} x^{-n} = \sum_{n=1}^{\infty}(2^{n-1}-1)x^{-n}.$$

The $n = 1$ coefficient is $2^0 - 1 = 0$, so the series really starts at x^{-2}:

$$\boxed{f(x) = \sum_{n=2}^{\infty} (2^{n-1} - 1)x^{-n} = \frac{1}{x^2} + \frac{3}{x^3} + \frac{7}{x^4} + \frac{15}{x^5} + \cdots \qquad (|x| > 2).}$$

(Combining the two series term by term is legitimate because each converges.) Either the two-series form or the combined form is a correct answer.

Checks

- *Leading behaviour.* For large $|x|$, $f(x) = \frac{1}{x^2 - 3x + 2} \approx \frac{1}{x^2}$, which matches the first term of the last expansion. The vanishing x^{-1} coefficient is no accident: the two partial fractions have equal and opposite $\frac{1}{x}$ behaviour.
- *A value.* At $x = 3$, $f(3) = \frac{1}{2 \cdot 1} = \frac{1}{2}$, and $\sum_{n \geq 2} (2^{n-1} - 1)3^{-n} = \frac{1}{2} \sum_{n \geq 2} (\frac{2}{3})^n - \sum_{n \geq 2} (\frac{1}{3})^n = \frac{1}{2} \cdot \frac{4}{3} - \frac{1}{6} = \frac{1}{2}$. Similarly $f(\frac{1}{2}) = \frac{4}{3}$ and $f(\frac{3}{2}) = -4$ agree with the first two expansions.
- Numerically, all three series (summed to 4000 terms) agree with f to about 10^{-14} at hundreds of random points in their ranges, positive and negative.

Common slips

- Getting the sign wrong in $\frac{1}{x-2} = -\frac{1}{2} \cdot \frac{1}{1 - x/2}$: the factor pulled out is -2, not 2.
- Being out by one in the index after multiplying by $\frac{1}{x}$: $\frac{1}{x} \sum_{n \geq 0} x^{-n}$ starts at x^{-1}, so the relabelled sum starts at $n = 1$.
- Using the same expansion in every range without checking that the ratio has modulus less than 1.

Beyond STEP

These three expansions are the real-variable shadow of *Laurent series* in complex analysis. For complex z, the function $\frac{1}{(z-1)(z-2)}$ has a different Laurent expansion in each of the regions $|z| < 1$, $1 < |z| < 2$ and $|z| > 2$: a disc and two annuli bounded by circles through the poles, which is exactly what the circles $|x| = 1$ and $|x| = 2$ become. The coefficients of z^{-1} in such expansions (residues) turn out to control contour integrals. Needham's *Visual Complex Analysis* gives a beautifully geometric account of power and Laurent series.

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